MIL-C-1311E
17 May 1972
SUPERSEDING
MIL-C-1311D
23 January 1967

MILITARY SPECIFICATION

CARTRIDGE, CALIBER .45, BALL, M1911

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, Caliber .45, Ball, M1911 intended for use in Caliber .45 weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition Of

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition Through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1168	- Lot Numbering of Ammunition

FSC 1305

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2.1 (Cont'd)

DRAWINGS

U. S. Army Munitions Command

C11738401	- Cartridge, Caliber .45, Ball, M1911
C11738402	- Cartridge, Caliber .45, Ball, M1911 (Steel Case)
D7553728	- Packing and Marking for Cartridges, Caliber .45; Cartons, Box, Ammunition, M2A1; Box, Wirebound
C7643674	-Classification of Cartridge Case Defects

PUBLICATIONS

Munitions Command

AMCR 715-505, Volume 5 - Ammunition Ballistic Acceptance Test Methods, Volume 5, Test Procedures for Cal. .45 Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with the requirements specified on Drawing C11738401 or C11738402 as applicable and as specified herein. (See 6.1.2).

3.2 Bullet extraction.-The force required to extract the bullet from the cartridge case shall be not less than 40 pounds.

3.3 Residual stress.-(Brass case cartridges) The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Velocity.-The average velocity of the sample cartridges, conditioned at 70° ± 2° Fahrenheit (F), shall be 855 feet per second (ft/sec) plus or minus 25 ft/sec, at 25.5 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 27 ft/sec.

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3.5 Chamber pressure.-The average chamber pressure of the sample cartridges, conditioned at 70° ± 2°F, shall not exceed 19,000 pounds per square inch (psi).

3.6 Accuracy.-The average of the diagonals ($\bar{D}$) of all targets of the sample cartridges fired at 50 yards shall not exceed 7.46 inches.

3.7 Waterproof.-The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 psi for 15 seconds.

3.8 Salt spray (steel case cartridges).-No steel case shall show any corrosion products after the cartridge has been subjected to a five percent salt spray for five hours.

3.9 Function and casualty.-The cartridge shall function without casualty.

3.10 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects.-The cartridge shall be free of wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.10.2 Foreign matter.-The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 1600 cartridges. The sample shall be manufactured using

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4.2.1 (Cont'd)

the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval. Determination as to acceptability of any initial production sample shall be based upon results of initial tests only and no second tests shall be permitted on that inital production sample.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.-Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number in accordance with MIL-STD-1168. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.-One hundred per cent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 per cent and the AQL for the minor class shall be 1.50 per cent.

4.3.2.1 Classification of defects.-The classification of defects shall be specified in Table I.

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Table I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case				
	in K, L, or M location	X			
	in I, S, or J location		X		
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
11	Scaly metal				X
12	No chamfer or head (rim)				X
13	Wrinkle			X	
14	Bulge			X	
15	Illegible or missing head stamp			X	
16	Defective head			X	
17	Defective mouth			X	
17a	Defective protective finish (steel case)			X	
	Bullet				
18	Dent			X	
19	Scratch			X	
20	Split bullet jacket		X		
21	Loose bullet		X		
22	Scaly metal (bullet)				X
23	Upset (crooked) point			X	
24	Exposed steel (clad jacket)			X	
25	Flat point			X	
	Primer				
26	No primer	X			
27	Cocked primer	X			
28	Inverted primer	X			
29	Loose primer		X		
30	Nicked or dented primer			X	
31	No waterproofing material (primer pocket joint)			X	

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Table I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Gaging				
32	Cartridge profile failure (requiring more than 10 lbs. dead weight to insertin profile and alignment gage)		X		
33	Diameter of extractor groove			X	
34	Diameter of head			X	
35	Thickness of head, max.			X	
36	Thickness of head, min.		X		
37	Case length		X		
38	Total length		X		
39	Depth of primer		X		

1/ Refer to MIL-STD-636 for visual standards for defects 1 thru 31. In the event of conflict between Table I of this specification and MIL-STD-636, as to defect classification, the classification specified in Table I shall apply.

4.3.3 Tests.-The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.-The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into samples for the various tests.

Table II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	20	3.2
Residual stress (mercurous nitrate) 1/	20	3.3

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TABLE II (Cont'd)

Test	Number of Cartridges	Requirement Paragraph
Velocity 2/	20	3.4
Chamber pressure 2/ 5/		3.5
Accuracy 2/	90	3.6
Waterproof 3/	20	3.7
Salt Spray (steel only) 2/	20	3.8
Function and casualty 4/		3.9
Pistol, automatic, cal. .45, M1911A1	294	
Gun, submachine, cal. .45, M3A1	900	

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges in the first sample. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

3/ Failure of nine or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than four, but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. The lot shall be rejected if, in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

4/ The lot shall be rejected when function and casualty defects, plus firing defects observed in all other firing tests (excluding erosion tests) exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in all the service weapons specified therefor.

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4.3.3.1 (Cont'd)

This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

5/ Velocity and pressure are obtained simultaneously.

4.3.3.2 Firing defects.-Firing defects and acceptance numbers shall be as specified in Table III.

Table III

Defects	Acceptance Number First Sample	Acceptance Number Cumulative (First & Second Samples)
Misfire 5/		
a. No vent hole, or obstruction in the vent area	0	-
b. Other	1	2
Bullet remaining in bore 1/	0	-
Primer leak		
a. Perforation in firing pin indent in primer cup		
(1) Submachine gun	14	41
(2) Automatic pistol	1	3
b. Escape of gas around primer cup 2/	14	38
c. Primer remains in pocket, but is physically loose	2	4
d. Blown primer	0	1
e. Primer falls out of pocket on retraction of bolt	0	1
Case casualty		
a. Longitudinal split 3/		
(1) Mouth (I)	27	75
(2) Body (J)	7	19

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Table III (Cont'd)

Defects	Acceptance Number: First Sample	Acceptance Number: Cumulative (First & Second Samples)
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 3/		
(1) Partial (J)	6	15
(2) Partial body (K)	0	1
(3) Partial head (L)	0	1
(4) Complete	0	1
Failure to extract	0	1
Weapon stoppage 4/	0	1

1/ No second sample permitted. Lot shall be rejected.

2/ Gas escapes around more than fifty per cent of periphery of cup.

3/ For location of defects indicated by letters in parenthesis, see drawing C7643674.

4/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

5/ Each cartridge that misfires shall be disassembled and examined for presence of a vent hole in the primer pocket, or any obstruction in the vent hole area of the primer pocket that can be assignable as the cause for misfire. If the vent hole is missing or obstructed, the lot shall be rejected with no second sample permitted.

4.3.4 Packaging, packing and marking inspection. During or immediately prior to the packaging operation, 100 per cent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy or blank cartridge shall be classed as a critical defect. Occurrence of any type other than those listed shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

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4.3.5 Inspection equipment.-Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. Simulated assessment of reference cartridges shall be made in accordance with AMCR 715-505, Volume 5.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.-The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three inches nor more than six inches per minute. The test shall be conducted in accordance with AMCR 715-505, Volume 5.

4.4.2 Residual stress (mercurous nitrate).-Tests shall be conducted in accordance with AMCR 715-505, Volume 5.

4.4.3 Velocity.-The tests shall be conducted in accordance with AMCR 715-505, Volume 5 and shall be conducted simultaneously with the chamber pressure tests.

4.4.4 Chamber pressure.-The tests shall be conducted in accordance with AMCR 715-505, Volume 5.

4.4.5 Accuracy.-The tests shall be conducted in accordance with AMCR 715-505, Volume 5.

4.4.6 Waterproof.-The test shall be conducted in accordance with AMCR 715-505, Volume 5. The container shall be evacuated to a pressure of 5 pounds per square inch (10 inches of mercury) below atmospheric pressure and held at that pressure for 15 seconds.

4.4.7 Salt spray.-Steel case cartridges shall be subjected to the salt spray test conducted in accordance with Federal Test Method Standard No. 151, in a five per cent solution, with the cartridges suspended bullet down for five hours.

4.4.8 Function and casualty.-The test shall be conducted in accordance with AMCR 715-505, Volume 5 and in the quantities listed below. The weapons shall be at room temperature at the beginning of test.

a. Pistol automatic, Caliber .45, M1911A1:
One hundred and forty-seven cartridges (twenty-one magazines, seven cartridges each) in each of two pistols.

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4.4.8 (Cont'd)

b. Gun, Submachine, Caliber .45, M3A1
Three hundred cartridges (ten magazines, thirty cartridges each) in each of three guns.

4.4.9 Defect penalty.-In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance of initial production sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level A.-(Worldwide shipment) - The cartridge shall be packed in accordance with Drawing D7553728.

5.2 Marking and labeling.-Packing boxes shall be marked and labeled in accordance with drawing cited in 5.1.

6. NOTES

6.1 Ordering data.-Invitation for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Applicable cartridge drawing number (See 3.1).

6.1.3 Type and level of packing.

6.1.4 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.5 Provision for the submission of acceptance inspection parts containing final inspection results for each lot of ammunition presented to the Government.

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6.1.6 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

Custodians:

Army - MU

Air Force - 70

User activity:

Navy - MC

Review activities:

Army - MU

Air Force - 70

Preparing activity:

Army - MU

Project No. 1305-0702

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

5. SUBMITTED BY *(Print or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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